PROJECT SUMMARY

1. Project Overview: Coronary Artery Segmentation & Stenosis Quantification

What Was Done:

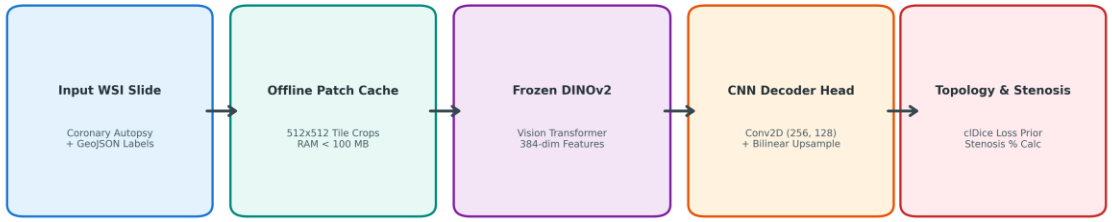
✔ **Objective:** Automated segmentation of concentric coronary artery layers (Media, Intima, Lumen) from autopsy histology slides.

✔ **Clinical Goal:** Automatically compute plaque stenosis % to measure arterial narrowing without manual tracing.

✔ **Memory Problem Solved:** Raw slides are >15,000x15,000 pixels (16 GB crash). Cropped artery ROIs and cached 512x512 tiles offline (RAM < 100 MB, epoch time < 2 min).

✔ **Connectivity Problem Solved:** Standard loss produced broken vessel rings. Applied clDice loss to enforce 100% closed, continuous wall boundaries.

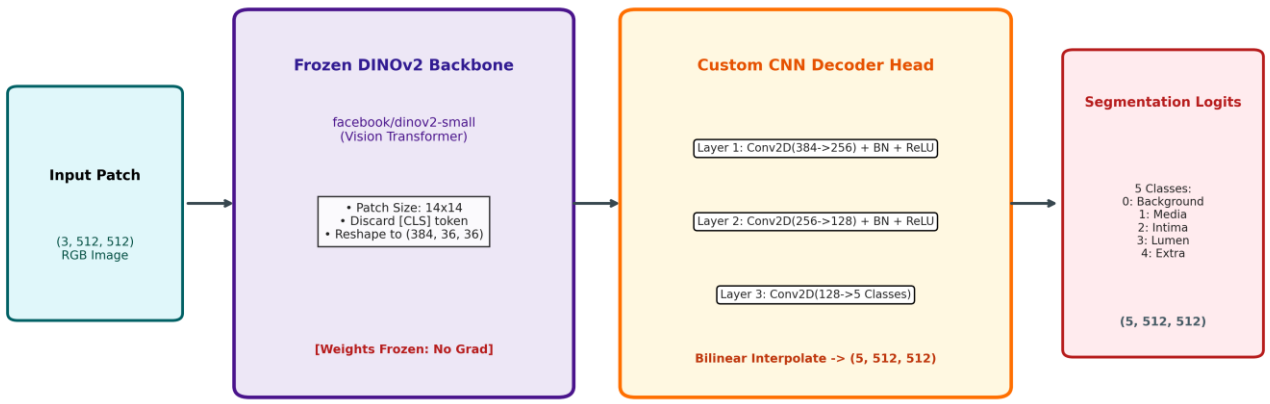
Figure 3.1: End-to-End System Pipeline for Coronary Artery Segmentation & Stenosis Quantification



PROJECT SUMMARY

2. Methodology: Frozen DINOv2 Backbone + CNN Decoder + clDice Loss

Figure 3.2: Detailed Neural Network Architecture (Frozen DINOv2 Feature Extractor + CNN Decoder)

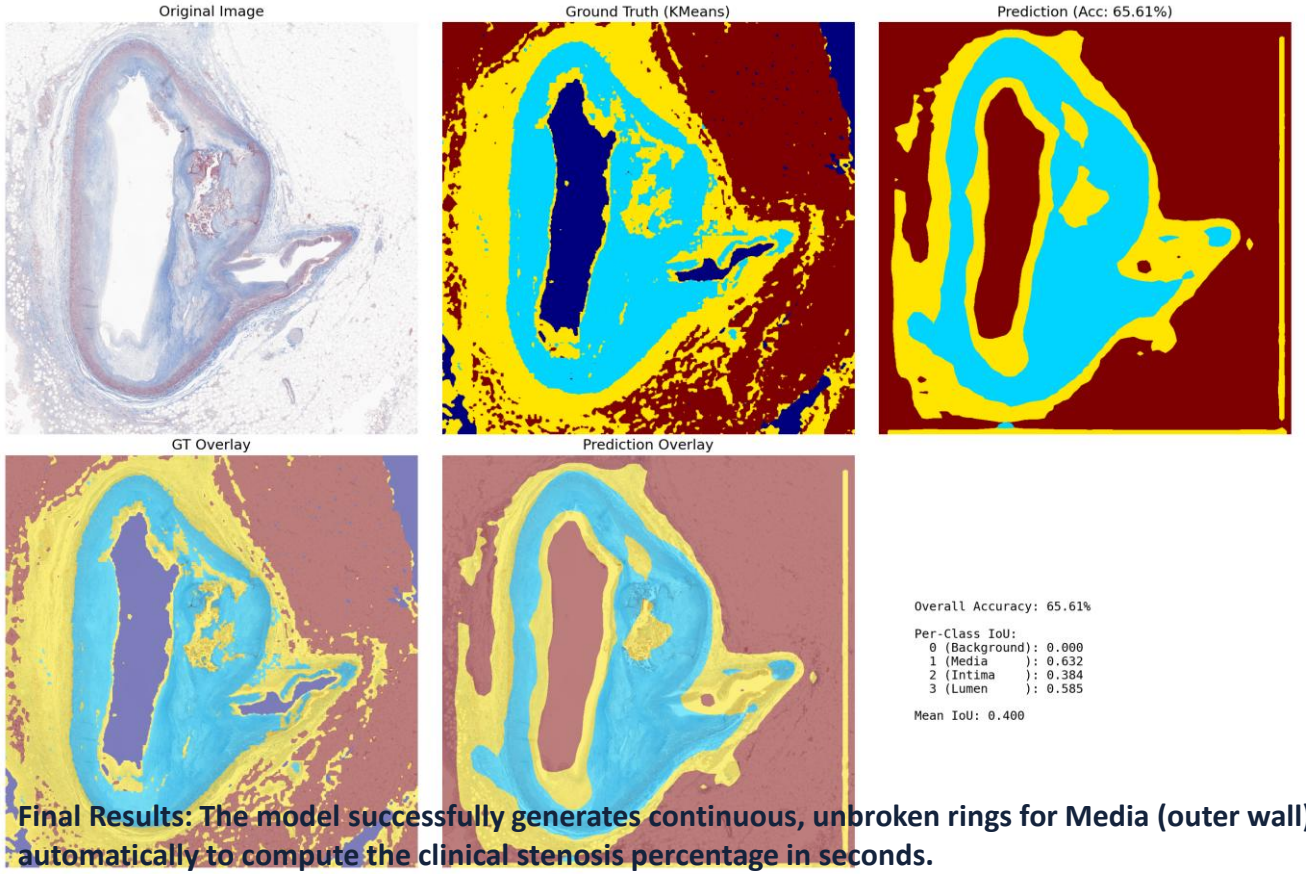


Methodology Breakdown:

- **Frozen DINOv2 Backbone:** facebook/dinov2-small Vision Transformer. Weights frozen (no grad) to eliminate overfitting on 16 autopsy slides.
- **Token Reshaping:** Extracts 14x14 patch tokens, discards [CLS], and reshapes into 384-channel 2D feature maps (384x36x36).
- **3-Layer CNN Decoder:** Conv2D (384->256) + BN + ReLU -> Conv2D (256->128) + BN + ReLU -> Conv2D (128->5 classes) + Bilinear Upsampling to 512x512.
- **clDice Loss:** Uses soft-erosion pooling to extract a 1-pixel centerline skeleton and penalizes broken rings during training.

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3. Final Output: Unbroken Segmentations & Automated Stenosis Quantification

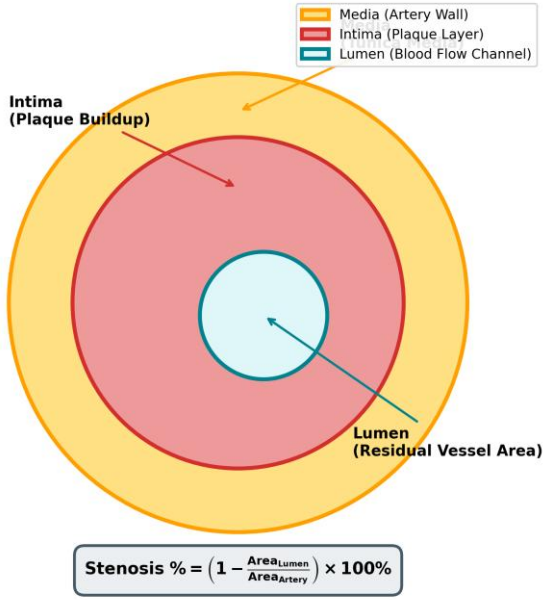



```
Overall Accuracy: 65.61%

Per-Class IoU:
  0 (Background): 0.000
  1 (Media     ): 0.632
  2 (Intima    ): 0.384
  3 (Lumen     ): 0.585

Mean IoU: 0.400
```

Figure 3.4: Concentric Coronary Tissue Layers and Clinical Stenosis Measurement Geometry



$$\text{Stenosis \%} = \left(1 - \frac{\text{Area}_{\text{Lumen}}}{\text{Area}_{\text{Artery}}}\right) \times 100\%$$

Final Results: The model successfully generates continuous, unbroken rings for Media (outer wall) and Lumen (blood channel). Surface areas are extracted automatically to compute the clinical stenosis percentage in seconds.